Statistical Decision Making
45-752
A/B Testing

Topics Covered in This Slide Deck:

Designing an Experiment To Test A Hypothesis About a Population Mean

Designing an Experiment To Test A Hypothesis About a Difference in Population Means

This file is named 02_2_slides.pptx

The accompanying Python commands are near the end of 02_notebook.ipynb

1

Introduction to This Slide Deck

In the previous slide deck, we discussed the logic and approach to A/B testing.

- We saw how test a difference in means by hand, and by using regression.
- A key issue in designing an experiment is determining how large a sample is needed.
- In this slide deck, we will develop the conceptual framework for designing an experiment.
- We will employ the approach to determine samle size for an experiment to test a hypothesis about a population mean.
- We will then employ the approach to study how to determine sizes for an experiment to test a hypothesis about a difference in population means.

2

2

Topics In This Slide Deck

Experimental Design: Conceptual Framework

Experimental Design: Choosing Sample Size For An Experiment to Test A Hypothesis about a Population Mean

Experimental Design: Choosing Sample Size For An Experiment to Test A Difference in Population Means

Experimental Design Application: Are Opioids More Effective Than non-Opioids for Pain Relief?

Experiment: Management Consulting in India

Summary of Topics Thus Far In Our Course

3

3

Statistical Decision Making
45-752

The Power of A Hypothesis Test

4

Four Possible Outcomes of A Hypothesis Test

		Null Hypothesis	
		True	False
Decision	Accept	Correct Decision Probability= (1-**α**)	Type II Error Probability = **β**
	Reject	Type I Error Probability= **α** Called Significance Level	Correct Decision Probability= (1-**β**) Called Power

5

5

Experiment Planned by Automotive Testing Service

Automobile Testing Service (ATS) is a company that provides independent evaluation of fuel economy claims.

An automobile manufacturer claims its new hybrid SUV will obtain 40 miles per gallon (MPG) in normal city driving.

Automobile Testing Service (ATS) is undertaking evaluation of this fuel economy claim.

- ATS has designed a route in a major metropolitan area that is typical of city driving.
- ATS plans to select a random sample of n=50 drivers from a group of licensed volunteers.
- Each driver will drive a randomly chosen SUV from the manufacturer's inventory.
- ATS will then assess whether the mean mpg from the sample, $\bar{X}$, is or is not consistent with the manufacturer's claim.

6

6

Calculating Power for ATS

Before conducting its evaluation, ATS wants us to calculate the power of the test it is planning to conduct.

Information and notation for the power calculation for the planned experiment:

- μ is the unknown true MPG of this SUV
- μ_0=40 is the MPG claimed by the manufacturer
- Sample size n=50
- Based on prior experience, ATS expects σ=5 for this SUV.
- ATS chooses significance level $\alpha = .05$
- ATS asks us to calculate the power of its test if the manufacturer's claim is wrong by 2 mpg.

7

7

Experimental Design – Calculating Power

Steps in the power calculation:

Step 1: Form the acceptance interval assuming the manufacturers claim is correct: $\mu = \mu_0 = 40$

Step 2: Calculate the probability of rejecting the null hypothesis if the difference between null and alternative is 2 mpg (i.e, the true mpg is either μ = 38 or μ = 42).

We will calculate power for μ = 42. We will then see that we get the same result if μ = 38.

8

8

Illustration of Power Calculation: Step 1

TEPPER

Form acceptance interval

$$[\mu_0 - 1.96\frac{\sigma}{\sqrt{n}}, \mu_0 + 1.96\frac{\sigma}{\sqrt{n}}]$$

With μ_0=40, σ=5, and n=50.

[40-1.96*5/50^.5, 40+1.96*5/50^.5]

= [38.614,41.386]

If the null hypothesis is true, there is a 95% probability that the sample average Miles Per Gallon obtained by ATS will be in this interval.

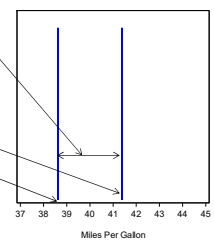
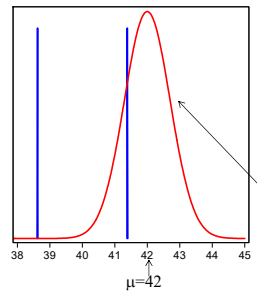


Illustration of Power Calculation: Step 2

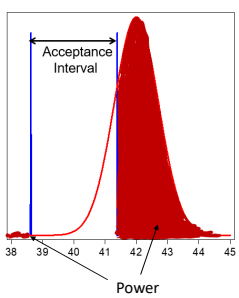
TEPPER



- ATS asks us to calculate power if true MPG differs from manufacturer's claim by 2 MPG.
- The sample mean $\bar{X}$ is a random variable.
- ATS will obtain one sample mean $\bar{X}$ when it conducts is evaluation.
- Here is the plot of the distribution of possible values of $\bar{X}$ ATS might obtain if the true MPG is μ=42 rather than 40, and the standard deviation is 5.

Illustration of Power Calculation Continued

TEPPER



The acceptance interval is centered on the null value μ_0=40 shown by vertical lines in the plot.

The red distribution curve is the distribution of $\bar{X}$ if μ=42 (and σ=5).

The red area under the distribution curve and outside the acceptance interval is the probability you reject the null hypothesis μ_0=40 if the true MPG is μ=42.

Hence, the sum of the red areas is the Power of the test when the true mean mpg is μ=42 and the null hypothesis is μ_0=40.

We can use the normal distribution to calculate power because n is large enough to rely on the central limit theorem.

Optional Detail for Power Calculation

Power Calculation Using Python

TEPPER

Information for our hypothesis test:

- Sample size = 50
- $\alpha = .05$
- Effect Size = 2
- Standard deviation, σ=5

```
# statsmodels.stats.power.tt_solve_power¶
# solve for any one parameter of the power of a one sample t-test

# Test of null hypothesis for a single mean: Calculate power with nobs=50
effect_size = 2
stdev = 5
standardized_effect_size = effect_size / stdev
stats.power.tt_solve_power(effect_size= standardized_effect_size, nobs=50, alpha=0.05)
```

The result is on the next slide

Power Calculation Using Python

We obtain the following output from Python:

np.float64(0.7917871892678123)

- Hence, we see that we obtain power = 0.79. There is a 79% probability that ATS will reject the manufacturer's claim if the true MPG is 42.
  - Rejecting 40 because the true MPG is 42 would be good news for the manufacturer because fuel economy would be higher than the manufacturer claimed.
- If we test true mean=38 instead of 42, we have the same effect size and hence will obtain the same power. Please verify this by running the command on the previous slide with effect size -2 instead of 2.

13

13

Summary

Calculating power for test of a hypothesis about a population mean requires us to specify four items:

- Difference between null and alternative hypothesis
- Standard deviation of the population being studied
- Sample size
- Significance level

In the next slides, we will see how power is used in design of A/B experiments.

14

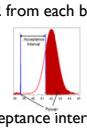
14

Optional: How Is Power Calculated

Here is how power is calculated for the ATS example.

- We calculated the acceptance interval: [38.614,41.386]
- Standardize these acceptance bounds around μ=42 by subtracting 42 from each bound and dividing by $\sigma/\sqrt{n}$ as follows:

$$\left[\frac{(38.614-42)}{5/\sqrt{50}}, \frac{(41.386-42)}{5/\sqrt{50}}\right]$$



- This yields the standardized bounds: [-4.788,-.8683]

Using these standardized bounds, we calculate the areas outside the acceptance interval. These areas are illustrated in solid red on the graph showing power.

- Small red area on the left is barely above zero.

```
from scipy.stats import norm
norm.cdf(-4.788,0,1)

np.float64(8.422582429567219e-07)
```

- Larger area on the right is .8073:

```
from scipy.stats import norm
1-norm.cdf(-.8683,0,1)

np.float64(0.807384938909162)
```

Adding the above two areas, we obtain power .81. There is an 81% probability that ATS will reject the Manufacturer's claim if the actual MPG differs from the claim by 2 MPG.

Note the slight difference from result on earlier slide is round off error. Back

15

15

Statistical Decision Making
45-752

Experimental Design:
Choosing Sample Size For An
Experiment to Test a Population Mean

For further reading on this topic see:
Chapter05_Experimental_Design.pdf

16

Sample Size and Power

In our previous slides (and optional video), we saw how to calculate power for an experiment planned by Automotive Testing Service (ATS).

We saw that we required the following information to calculate power:

- Standard deviation of the population being studied
- Significance level
- Difference between null and alternative hypothesis
- Sample size

In designing an experiment, we must choose sample size.

To do so, we must specify the first three items above and also specify the desired level of power. We then calculate the require sample size.

17

17

Calculating Sample Size To Test A Hypothesis About a Mean

Let's calculate the sample size required by ATS for the SUV we have been studying if ATS wishes power=.85. As before:

- ATS estimates the standard deviation = 5
- ATS chooses significance level α to be .05
- ATS specifies a Difference between null and alternative hypothesis of $|\mu-\mu_0|=2$ miles per gallon. This difference is called the Effect Size of the experiment.
- ATS chooses power = .85

18

18

Building Intuition about Effect Size

We introduced the concept of Effect Size on the previous slide. We now consider examples to build intuition about Effect Size.

Consider the ATS experiment to evaluate SUV mileage.

- Suppose the true mpg of the SUV being evaluated is 15mpg.
- Would it be hard to reject the manufacturer's claim of 40 mpg?
- No, it would be easy.
- Why would it be easy?
- A relatively small sample would be sufficient to reject the claim of 40 mpg because a difference (effect size) of 25 mpg between null (40) and true (15) would be easy to detect.

19

19

Building Intuition about Effect Size Cont'd

- Suppose the true mpg of the SUV is 39.5 mpg.
- Would it be hard to reject the manufacturer's claim of 40 mpg?
- Yes.
- Why would it be hard?
- A very large sample would be required to reject the claim of 40 mpg because such a small difference (effect size) of 0.5 mpg between null (40) and true (39.5) would be very difficult to detect.

20

20

Building Intuition about Effect Size Cont'd

- Suppose the true mpg of the SUV is 15 mpg.
  - Would it be important for Automotive Testing Service to reject the manufacturer's claim of 40 mpg?
  - Duh!
  - This would obviously be important!
- Suppose the true mpg of the SUV is 39.5.
  - Would it be important for ATS to reject the manufacturer's claim of 40 mpg?
  - Probably not. Such a small deviation from the actual mpg is not of much concern.
- **Effect Size:** As a manager of ATS, you need to decide **how small a difference you will seek to detect between the null and alternative** with your experiment.

21

21

As Manager of ATS You Must Answer Four Questions

Q1. What effect size is important for you to detect?

You decide that a deviation of 2mpg or more from the manufacturer's claim is important to detect. Hence, you choose an **Effect Size** = 2 MPG.

Q2. How certain do you want to be that the experiment will detect a 2 MPG deviation?

You choose 85%. (Remember that this is called the **Power** of the test.)

Q3. What **Significance Level** do you want for your test?

You choose **α** = .05.

Q4. What is the **Standard Deviation** of MPG for this vehicle?

Based on your prior experience at ATS, you estimate the standard deviation will be about 5 MPG.

Having made the above tentative decisions, you can calculate the necessary sample size. You may reconsider your decisions based on the cost of the sample size that you find in your calculations.

22

22

Using Python to Calculate Sample Size

The information for our calculation is:

- Power (we chose .85)
- $\alpha = .05$
- Effect size = 2
- Standard deviation, σ=5

```
# Calculate sample size, Automotive Testing Service, test null hypothesis for a
# single mean
effect_size = 2
stdev = 5
standardized_effect_size = effect_size / stdev
stats.power.tt_solve_power(effect_size=standardized_effect_size, alpha=0.05,powe
```

We obtain 58.067. Hence, ATS will need 58 cars.

23

23

Statistical Decision Making
45-752

Experimental Design:
Choosing Sample Size For An Experiment
to Test A Difference in Population Means

For further reading on this topic see:
Chapter05_Experimental_Design.pdf

24

Sample Size for A/B Testing

We have seen how to calculate sample size to test a hypothesis about a population mean.

Much more common is testing a difference in population means.

Suppose ATS wishes to compare the mileage of two different subcompact cars, A and B.

- The null hypothesis is that they are equally fuel efficient.
- An Effect Size of 2 MPG is chosen by ATS.
- The standard deviation of MPG for each vehicle is 3
- Significance level = .05
- Power = .85

25

25

Calculating Sample Size for Test of Difference in Means

Enter the information from the previous slide for the effect size (2), the standard deviation for each vehicle (3), the significance level (.05), the power (.85), and the ratio of the number of observations in sample 2 relative to sample 1. We specify ratio 1 because we want the same number of cars of each type.

```
# ATS sample size to test difference in population means
# Desired power of the test is .85

tprstats.AB_nobs(diff_means=2, std1= 3, std2=3, alpha=0.05, power = 0.85, ratio=1)
```

We obtain 41.34 cars per sample. Rounding, we conclude that ATS needs 41 vehicles of each type.

- A more conservative approach would be to round up to 42.
- In the above command, we set ratio= 1.
- If it is more costly to test one type of vehicle than another, we might want to use more of the less expensive vehicle and fewer of the expensive one.

26

26

Calculating Sample Size for Test of Difference in Means Continued

Suppose it is more costly to obtain type B vehicles to test than it is to obtain type A vehicles. ATS might then set ratio=4/6.

The command to calculate sample sizes is then:

tprstats.AB_nobs(diff_means=2, std1= 3, std2=3, alpha=0.05, power = 0.85, ratio=4/6)

From this command, we obtain a sample size of 86.47 with 34.45 Type B vehicles and 51.68 Type A vehicles.

Hence, rounding, we obtain 35 vehicles of Type B and 52 of Type A.

- Thus, a total of 87 vehicles need to be tested.
- With unequal sample sizes, the total number of vehicles to be tested is higher (87) than with equal samples sizes (82). When the standard deviations are the same for the two types, unequal sample sizes will always require a larger total.

Note: We want both samples to be large enough to satisfy the central limit theorem, so, with unequal samples, we typically want to require that each sample have 30 or more.

27

27

Costs of Experimentation

In some situations, increasing sample size can be relatively inexpensive.

- Everyday business operations may provide relatively low cost opportunities to obtain large samples.
- For example, we previously discussed the extensive use of A/B testing in online settings.

In some cases, increasing sample size is very expensive.

A study of 225 individual clinical trials for new medications found:

- The median cost of the trials was $19 million with the 25th and 75th quantiles being $12 million and $33 million respectively.
- The median cost per patient in a trial was $41,413 with the 25th and 75th quantiles being $29,894 and $75,047 respectively.

Source: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7295430/

28

28

Tradeoffs You Face As A Manager

- Increasing your sample size increases the cost of your experiment, with cost varying depending on the situation being studied.
- The smaller the Effect Size you choose, the bigger your sample must be.
- The more power you want from your test, the bigger your sample must be.
- The smaller the significance level, α, the larger your sample must be.
- ➢ Be sure you understand why each of the above is true.
- To solidify your understanding, you might find it useful to do calculations using the ATS example, varying one-at-a-time effect size, power, and significance level.

29

29

Statistical Decision Making
45-752

Experimental Design:
Are Opioids More Effective Than non-Opioids for Pain Relief?

30

Calculating Sample Size: Another Application

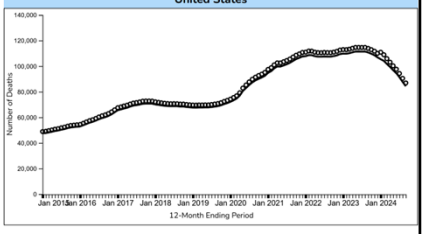
News reports on the problem of Opioid abuse declined with the emergence of Covid.

Unfortunately, the scourge of opioid abuse has continued. The graph to the right shows overdose deaths by year from 2013 to 2024.

If there is anything slightly positive in this, it is that the number of overdose deaths seems to be declining.

Source: https://www.cdc.gov/nchs/nvss/vsrr/drug-overdose-data.htm

Figure 1a. 12 Month-ending Provisional Counts of Drug Overdose Deaths: United States



- Each dot on the graph shows the numbers of deaths due to drug overdose in the 12-month period ending in the month indicated.
- In the 12-month period ending June 2025, there were 72,877 overdose deaths.

31

31

Costs of The Opioid Epidemic

While the toll on human health and welfare is the major cost of opioid abuse, the impacts on business and the economy are high—and rising:

- https://www.cbsnews.com/news/opioid-addiction-2-6-billion-business-cost/
- https://www.nytimes.com/2018/09/21/business/economy/opioid-overdose-workplace.html

A Committee of the US Congress estimated that: "The Economic Toll of the Opioid Crisis Reached Nearly $1.5 Trillion in 2020."

- https://www.jec.senate.gov/public/index.cfm/democrats/issue-briefs?ID=CE55E977-B473-414F-8B88-53EB55EB7C7C

Prescription of opioids for pain relief has been widely cited as a major factor leading to opioid abuse:

- https://www.unodc.org/unodc/en/frontpage/2018/June/world-drug-report-2018_-opioid-crisis--prescription-drug-abuse-expands-cocaine-and-opium-hit-record-highs.html
- https://www.drugabuse.gov/drugs-abuse/opioids/opioid-overdose-crisis

Are opioids really more effective than safe, non-addictive alternatives for pain relief? Two recent experiments have studied this question.

32

32

A/B Test of Opioids vs Non-Opioids

On March 6, 2018 Los Angeles Times reported:

"In the first randomized clinical trial to make a head-to-head comparison between opioids and other kinds of pain medications, patients who took opioids fared no better over the long term than patients who used safer alternatives."

- All of the patients had pain in their backs, hips or knees for at least six months, and that pain was bad enough to interfere with their daily activities and enjoyment of life.
- The study randomly assigned 119 individuals to receive opioids for their pain and 119 to receive non-opioids for their pain.

http://www.latimes.com/science/sciencenow/la-sci-sn-opioid-painkillers-no-better-20180306-story.html

33

33

Application: Design of The Experiment

We now study the design of the experiment comparing the effectiveness for pain relief of Nonopiod relative to Opioid pain medication.

By way of background: Pain intensity is measured as follows.

- A patient is asked to indicate pain intensity on an 11-point scale with 0 being no pain and 10 being the worst imaginable pain.
- This is called a Brief Pain Inventory (BPI) scale, and is shown below:

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

No Pain — Pain As Bad As You Can Imagine

34

34

Sample Size in The Opioid Experiment

Here is a copy of the authors report of their calculation of sample size:

> **Statistical Analysis**
> Assuming a 2-sided α level of .05 and a standard deviation of 2.7,[30] 115 patients completing the study per group were required for 80% power to detect a 1-point between-group difference in mean BPI interference at 12 months.[16]

https://jamanetwork.com/journals/jama/article-abstract/2673971?redirect=true

Let's denote the two treatments above as A and B.

- We see above that the authors chose an effect size=1.
- Based on their prior experience, they expected equal standard deviations of the following magnitude: σ_A=2.7 and σ_B=2.7
- They wanted an equal number of subjects in each of the two groups.

35

35

Experiment Design: Alternatives to Opiods for Pain Relief

How did the authors determine that they needed 115 patients in each group?

From the previous slide, we see they specified the following:

- Effect size=1.
- Standard deviation for each group = 2.7
- α =.05
- Power=.8
- The authors wanted groups of equal size. Hence ratio= 1

Our Python command is:

```
tprstats.AB_nobs(diff_means= 1, std1 =2.7, std2=2.7, alpha=0.05, power = 0.8, ratio= 1)
```

We obtain:

nobs1	115.404644
nobs2	115.404644
nobs_total	230.809289

To allow for possible dropouts from the study, the authors decided to use 120 per group instead of 115.

This is an excellent example of experimental design. Be sure you "get it"!

36

36

Results of The Opioid Experiment

There were 119 individuals in each group:

- Experimental outcomes: At 12 months, mean BPI severity was 4.0 in the opioid group (SD, 2.0) vs 3.5 in the non-opioid group (SD, 1.9). SD denotes standard deviation.
- Remember that lower BPI is better, so the above tells us that the non-opioid outcome was better than the opioid outcome.
- Let X denote the non-opioid medication and Y the opioid medication, then $\bar{X} = 3.5$ and $\bar{Y} = 4$
- Is this difference significant? The information provided above permits us to do a test of differences.
- We now calculate the t-statistic and p-value for this hypothesis test.

37

37

Experiment: Alternatives to Opioids for Pain Relief

With standard deviations of 2.0 and 1.9 for the two groups and $n_1=n_2=119$, the t-statistic for testing the difference in means is:

$$t = \frac{(\bar{X} - \bar{Y})}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} = \frac{3.5 - 4.0}{\sqrt{\frac{2^2}{119} + \frac{1.9^2}{119}}} = -1.98$$

Using the Python command: 2*t.cdf(-1.98,238-2)

We obtain a p-value=.049

Our significance level is α=.05.

The p-value is less than α. The hypothesis that opioids and non-opioids were equally effective in reducing pain is rejected. Non-opioids were **significantly better i**n reducing chronic pain.

For several other measures, opioids were again no better than non-opioids.

CONCLUSIONS AND RELEVANCE Treatment with opioids was not superior to treatment with nonopioid medications for improving pain-related function over 12 months. Results do not support initiation of opioid therapy for moderate to severe chronic back pain or hip or knee osteoarthritis pain.

38

Another Experiment on Opioids for Pain

Another experiment on opioids for pain relief was conducted in hospital emergency rooms. The study found that an ibuprofen-Tylenol combination was as effective as opioids:

https://www.nytimes.com/2017/11/08/well/live/alternatives-to-opioids-for-pain-relief.html

Thus, two experiments on two very different types of pain, chronic pain and pain due to injuries, find that opioids are no better than non-opioids.

39

39

Experiment: Management Consulting in India

You might find it interesting to read about an experiment on management consulting several years ago in India.

This experiment has many elements and, hence, it is too extensive for us to cover in the time that we have available.

The next slide provides the abstract to the paper that the authors wrote summarizing the experiment.

Here is a link to the full report:

https://core.ac.uk/download/pdf/6281741.pdf

40

40

Abstract from The Authors Report

Does Management Matter?

Nicholas Bloom, Benn Eifert, Aprajit Mahajan, David McKenzie, John Roberts

A long-standing question is whether differences in management practices across firms can explain differences in productivity, especially in developing countries where these spreads appear particularly large. To investigate this, we ran a management field experiment on large Indian textile firms. We provided free consulting on management practices to randomly chosen treatment plants and compared their performance to a set of control plants. We find that adopting these management practices raised productivity by 17% in the first year through improved quality and efficiency and reduced inventory, and within three years led to the opening of more production plants. Why had the firms not adopted these profitable practices previously? Our results suggest that informational barriers were the primary factor explaining this lack of adoption. Also, because reallocation across firms appeared to be constrained by limits on managerial time, competition had not forced badly managed firms to exit.

Bottom Line: Management Matters!

41

41

Summary of First Two Sets of Slides

- We Reviewed some Fundamental Statistics Concepts
  - Central Limit Theorem
  - Hypothesis Testing
- We Reviewed Statistical Quality Control
- We Reviewed Testing Difference in Means and Applied to Water Conservation Experiment
- We Reviewed Single Variable Regression
- We Saw How to Use Regression to Test Difference in Means and Illustrated with Water Conservation Application
- We defined Power and saw how to calculate power.
- We saw how to choose sample size for an experiment.
- We saw how to use Python to utilize the statistical methods we have developed.

42

42